PE3101P: Decision and Social Choice

Lecture Notes 7 — Harsanyi's aggregation theorem

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1 October 2026

1 Social choice

1.1 Decision Theory vs Social Choice Theory

In the first half of the semester, we looked at (normative) decision theory. We were there interested in how rationality requires us to choose in light of risk. We were imagining that we already had preferences to hand, so as far as the non-risky cases went, the hard work was already done. You can think of the difficult cases in decision theory like this: an option can be better in some states of nature, but worse in some others; and we want to know how these considerations weigh up together to produce an overall ranking of the alternative gambles.

In the second half of the semester, we're going to be studying social choice theory. Roughly speaking, this is about how to choose when the tough thing about our decision isn't so much that an option is worse in some *states of nature* but better in others, but rather we're dealing with cases where an option may be worse *for some people* but better for others. That is: decision theory deals with **risk**. Social choice theory deals with **trade-offs between people**.

The two areas are tightly structurally and substantively related. They are substantively related in the direction of decision theory having lessons for social choice theory, more than the other way around, at any rate. This is because when we are choosing for many people, those choices often involve risk *as well* as trade-offs between people. So an adequate theory of social choice—a way of telling us how to choose between options which affect many people and lead to trade-offs between them, given the facts about their preferences or their wellbeing—needs to also tell us how to deal with risk.

The lessons of decision theory apply equally well to the context of social choice under risk. As we shall see in this lecture, decision-theoretic principles can also have consequences for social choice even when there is no risk involved (though not without the mediation of other principles).

1.2 Persons and States of Nature

Decision theory and social choice theory are structurally closely related because, although we won't focus on this in the course, there is quite a tight analogy between the role played by states of nature in decision theory and the role played by individual people in social choice or moral value theory.

So far, in decision theory we've often seen tables like these:

Table 1: A choice between two gambles on a fair coin. The numbers are utilities.

	Heads	Tails
Option A	5	5
Option B	11	0

In social choice theory, we will sometimes see tables like these:

Table 2: A choice between two outcomes for Alex and Darius. The numbers are wellbeing levels.

	Alex	Darius
Option A	5	5
Option B	11	0



Figure 1: A visual representation of the differences between decision theory and social choice theory.

As you can hopefully guess from looking at the tables, the role played by different states of nature in decision theory is quite similar to the role played by different people in social choice theory. In decision theory, we had the statewise dominance principle: one gamble is better than another if it is better in all states of nature. In social choice theory, we have the Pareto principle: one social outcome is better than another if it is better for each person. In both cases, we typically characterise the objects of comparison—gambles or social outcomes—as functions from states of nature/people to utilities, outcomes under a given preference ordering, or levels of individual wellbeing. And the theorem we are going to see today, if we accept its assumptions, posits a kind of two-way connection between individual attitudes to risk and social attitudes to inequality. (Note that inequality, on the persons/states of nature analogy, can be thought of as the social-choice-theoretic counterpart of risk, wherein variation across states of nature is replaced by variation across people.)

But that's getting ahead of ourselves: we'll see more about that later. Let's see a bit more about social choice theory first.

2 From preference to wellbeing

2.1 Preference and social choice

Social choice theory, then, asks us to choose between options which affect many people, not just one. To do this, we have a certain stock of information available. In some cases, including sometimes for pragmatic reasons, we only know about people's ordinal preferences. That is, we only know *what* they prefer to *what*, but we don't necessarily know *how much* they prefer things, or have any other information like that.

Now, a person's preferences tell us how they will decide for themselves. And if a utility function represents their preferences, they prefer whatever has the higher expected utility.

But that doesn't tell us how to choose on behalf of several people, when the decision affects them differently. They can disagree about which option is best, and so there might be no option that is "best for everyone". We need a way to balance people's interests against each other. What we're going to look at this week is Harsanyi's aggregation theorem, which gives an argument for one way of doing that.[1]

2.2 Two kinds of disagreement

People can disagree in two quite different ways. Suppose a liberal and a social democrat both want higher economic growth. The liberal thinks it will come from lower taxes; the social democrat thinks it will come from raising taxes in order to invest more in education. They want the same thing, and they disagree only about how to get it. That disagreement comes from their having different credences about what will happen.

That isn't the kind of disagreement we're interested in here. We're interested in cases where people are *affected* differently by a decision: where one option makes one person better off and another person worse off. To keep the first kind of disagreement out of the picture, we'll judge everything from *one* set of credences. Whenever we talk about the probability of something, it's the same probability for everyone.

2.3 Choose larger rather than smaller benefits?

Here is a simple example. There are two people, Alex and Darius, and two options. The numbers in Table 3 are their wellbeing levels: how well off each of them would be under each option.

Table 3: Wellbeing levels.

	Option 1	Option 2
Alex	8	2
Darius	3	5

Option 1 is better for Alex, by 6. Option 2 is better for Darius, by 2. Which should we choose? One natural answer is that we should benefit someone by more, rather than someone else by less, and so choose Option 1.

Could we have got numbers like these from Alex and Darius's preferences alone, using the utility functions that represent their preferences? Have a think about what might go wrong before reading on.

2.4 Utilities can be stretched; wellbeing differences cannot

Recall from week 4 that a utility function representing someone's preferences is not unique: it is unique only up to "positive affine transformation". This means that if u represents someone's preferences, then so does $au + c$, for any number $a > 0$ and any number c. Multiplying by a stretches (or shrinks) the function, and adding c shifts it up or down. Neither changes which prospects have the higher expected utility, and so neither changes which preferences are represented.

So suppose we now read the numbers in Table 3 as utilities representing Alex and Darius's preferences. Alex's utilities for the two options differ by 6, and Darius's by 2. But we could just as well stretch Darius's utility function by a factor of 10, as in Table 4.

[1] Harsanyi, John C. (1955). "Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparisons of Utility". In: *Journal of Political Economy* 63.4, pp. 309–321. The optional reading for this week is Gustafsson, Johan E. and Kacper Kowalczyk (2025). "Harsanyi's Utilitarian Theorems without Tears". In: *An Introduction to Utilitarianism*. Ed. by Richard Yetter Chappell, Darius Meissner, and William MacAskill.

Table 4: The same preferences, with Darius's utility function stretched by a factor of 10.

	Option 1	Option 2
Alex's utility	8	2
Darius's utility	30	50

Darius's new utility function represents exactly the same preferences as the old one. But now the difference for Darius is 20, which is bigger than the difference for Alex.

The difference in *utility* depends on how much each person's function is stretched, and nothing stops us from stretching Alex's, or shrinking Darius's, as much as we like. So if all we look at is people's preferences, there is no way to tell which option makes more of a difference to whom. We have to bring in some other information. That's the main reason we will talk about wellbeing, rather than preferences.

2.5 We need comparisons of wellbeing

And we need to be able to make comparisons like these if we're going to say anything plausible about what's best overall. Consider Table 5.

Table 5: Wellbeing levels.

	Option 1	Option 2
Alex	100	0
Darius	4	5

Here the choice makes a massive difference to Alex and a tiny difference to Darius, and there is no reason to favour Darius. Overall, the best person to benefit is Alex. But unless we can compare Alex's wellbeing with Darius's, we can't say that.

Preferences are enough for decision theory, which is about how a single person should choose. But they are arguably not enough to decide what's best overall, when the interests of different people pull in different directions. (Deciding from people's preferences alone is still sometimes useful. That's what voting theory does, which we'll come to later in the semester.)

2.6 Preference vs betterness

So we will work with what is good for each person, rather than with their preferences. Instead of

$$X \text{ is at least as } \textit{preferred} \text{ by person } i \text{ as } Y,$$

we will talk about

$$X \text{ is at least as } \textit{good for} \text{ person } i \text{ as } Y,$$

which we write $X \succsim_i Y$. We write $X \sim_i Y$ when X and Y are equally good for i, and $X \succ_i Y$ when X is better for i than Y. The reason for the switch is that we need facts about wellbeing, and what's good for someone is a matter of their wellbeing, not their preferences. On its own, the switch won't fix the scale problem from section 2.4, because a betterness relation, like a preference relation, is represented by a utility function only up to stretching and shifting. But for now, we will ignore this problem. We will just imagine that we can say that if the individual betterness relation satisfies the axioms of expected utility theory, then utility equals wellbeing. We'll return to the problem in section 6.

Each person's betterness relation ranks *prospects*, not just outcomes. As in week 4, a prospect is a probability distribution over outcomes. So the betterness relation takes account of the risks and benefits that a prospect holds for each person. An outcome counts as a prospect too: the prospect that gives that outcome for sure.

3 Aggregation

3.1 The social welfare ordering

A *social welfare ordering* is something like a betterness relation for the whole of society: everyone who is affected by the decision. It ranks gambles. We write $X \succsim Y$ for "X is at least as good as Y, overall", $X \sim Y$ for "X and Y are equally good", and $X \succ Y$ for "X is better than Y". (There's no subscript, because these are judgements about everyone taken together, not about any one person.) As usual, $X \sim Y$ means that $X \succsim Y$ and $Y \succsim X$, and $X \succ Y$ means that $X \succsim Y$ but not $Y \succsim X$; and the same goes for each person's $\sim_i$ and $\succ_i$.

The social welfare ordering is an abstract device, and it can be read in several ways:

– what society *should* choose;

– what society *will* choose;

– which outcome *ought to be chosen*, morally;

– which outcome would be *morally best*.

We can talk about social welfare orderings without deciding which of these we mean. I'll mostly have the last in mind: which outcome is best, when the choice involves trade-offs between people.

3.2 Harsanyi's aggregation theorem

Harsanyi's aggregation theorem starts with some really plausible-looking principles. From them, it derives a conclusion that is far from impossible to believe, but is definitely controversial. With n people, numbered 1 to n, the conclusion is this:

Utilitarianism: The best outcome is the one with the greatest sum of wellbeing.[2] That is,

$$X \succsim Y \quad \text{iff} \quad w_1(X) + \cdots + w_n(X) \;\geq\; w_1(Y) + \cdots + w_n(Y),$$

where $w_i(X)$ is person i's wellbeing in outcome X, and "iff" is short for "if and only if".

3.3 Same total, equally good

Consider the two outcomes in Table 6.

Table 6: Two outcomes with the same total.

	Alex	Darius	Total
Unequal	10	0	10
Equal	5	5	10

The totals are the same, so according to Utilitarianism, Unequal and Equal are equally good. On the face of it, that's pretty implausible. So this is going to be the best kind of philosophical argument: one that starts with premises that are hard to reject and ends with a conclusion that's hard to accept.

The proof gets from Unequal to Equal one step at a time, and each step uses one or two of the premises. As we go through it, look out for the step you would resist.

[2]Strictly, the theorem's conclusion is about a sum of *utilities*, chosen to match wellbeing in a way that section 6 makes precise. Until then, you can read the numbers in the tables as wellbeing levels; but in the proof, they are each person's utilities.

3.4 The premises

There are five premises. The first is about outcomes, and to state it we need a definition.[3]

Definition 1 (Anonymous equivalence)**.** Outcomes X and Y are *anonymously equivalent* if the people in X and Y can be matched up with unique counterparts, so that each person in X has the same level of wellbeing as their counterpart in Y.

In other words, two outcomes are anonymously equivalent when you can get from one to the other just by switching around who gets which level of wellbeing. (A way of switching people around like this is called a *permutation*.) The same levels of wellbeing are there in both outcomes, and the same number of people are at each level; all that changes is *who* is at which level.

Writing outcomes as lists of wellbeing levels in the order (Alex, Darius), the outcomes $(10, 0)$ and $(0, 10)$ are anonymously equivalent: match Alex in the first with Darius in the second, and Darius in the first with Alex in the second. Each of them has the same wellbeing level as their counterpart. If we add Philip, and write outcomes in the order (Alex, Darius, Philip), then $(1, 2, 3)$ and $(2, 3, 1)$ are anonymously equivalent too: match Alex in the first with Philip in the second, Darius with Alex, and Philip with Darius.

Anonymity: If outcomes X and Y are anonymously equivalent, then $X \sim Y$.

So Anonymity says that $(10, 0) \sim (0, 10)$, and that $(1, 2, 3) \sim (2, 3, 1)$. The thought is that Alex and Darius matter equally, and that what's at stake for each of them is the same, so we should be indifferent about who gets what.

The remaining premises are about gambles. Here, as with the decision tables of week 1, we describe a gamble by saying which outcome it gives in each state of nature. The same gamble can be written out with whatever states we like: $(10, 0)$ written out with a coin, giving $(10, 0)$ however it lands, is still the gamble that gives $(10, 0)$ for sure.

Statewise Dominance: If gamble X is at least as good as gamble Y in every state of nature, then $X \succsim Y$. If, in addition, X is better than Y in some state of nature, then $X \succ Y$.

"X is at least as good as Y in a state" means that the outcome X gives in that state is at least as good, overall, as the outcome Y gives in that state. For example, suppose the government can prepare for a natural disaster, and preparing costs nothing (Table 7). If the disaster happens, preparing leads to less damage than not preparing. If it doesn't happen, everything goes on as normal either way. So preparing is at least as good as not preparing whether or not the disaster happens, and better if it does. By Statewise Dominance, preparing is better.

Table 7: Preparing for a disaster, at no cost.

	Disaster	No disaster
Prepare	less damage	as normal
Don't prepare	more damage	as normal

Ex Ante Pareto: If gamble X is at least as good for everyone as gamble Y, then $X \succsim Y$. If, in addition, X is better for someone than Y, then $X \succ Y$.

"At least as good for everyone" means that $X \succsim_i Y$ for every person i. The principle is called "ex ante" because it applies to gambles, and not just to outcomes: it is about how things look for each person *before* the uncertainty is resolved. "Better for everyone" here doesn't mean "guaranteed to turn out better for everyone". It means that the gamble gives each person a better balance of risks and benefits.

[3]The definition is from my own work, where it is stated for populations: "populations X and Y are *anonymously equivalent* whenever we can obtain one from the other by permuting the set of possible people, so that the people in X and Y are matched up with unique counterparts, and each person in X has the same level of wellbeing as their counterpart in Y" (Francis n.d.). Francis, Tomi (n.d.). "Anonymity, Infinity, and the Separateness of Persons". Unpublished manuscript.

There might be a risk that some people will be worse off; and it might even be guaranteed that some people will be worse off, and yet the balance of risks for each person might improve. Consider a case where a government decides to encourage the adoption of a new technology. As with virtually all economic changes, there might be winners and losers, but it might be very unclear who the winners and losers will be. Nevertheless, if the technology increases overall economic output to a sufficient extent, each person might be expectedly better off: they're far more likely to be one of the many winners than to be one of the few losers. Ex Ante Pareto says that it's better to choose a gamble where everyone is a winner *in expectation*, even if there will be some losers once the die is cast.

How do we decide what's better for each person? Using expected utility theory, from week 4.

Expected Utility: Each person's betterness relation satisfies the axioms of expected utility theory. So it can be represented by a utility function, in the expected-utility sense.

That is, by the expected utility theorem from week 4, each person i has a utility function such that one prospect is at least as good for i as another if and only if it has at least as great an expected utility for i. From here on, the numbers we assign to people are their utilities, from utility functions that have been fitted to one another using the facts about wellbeing, so that the same number means the same level of wellbeing for everyone. Section 6 explains exactly what that means, and under what conditions it can be done. Until then, you can think of the numbers as wellbeing levels; and since the same number means the same wellbeing, two outcomes whose numbers can be matched up person to person are anonymously equivalent, so Anonymity applies to these utility numbers.

Transitivity: If $X \succsim Y$ and $Y \succsim Z$, then $X \succsim Z$.

This is what chains the steps of the proof together.

We also assume that every outcome (every list of numbers, one per person) is possible, and so is every lottery over finitely many of those outcomes, so that the outcomes and gambles the proof needs are always available.

The five premises all look quite innocuous. But together, they imply Utilitarianism—arguably, at least. (We will see some disagreement with that interpretation of the theorem later.)

4 The proof

4.1 A simple case

We start with the case from Table 6, and show that Unequal and Equal are equally good. We'll use a fair coin, so Heads and Tails each have probability $\frac{1}{2}$.

First, write out Unequal as a gamble, using the coin: Alex gets 10 and Darius gets 0, however the coin lands. Now compare it with a second gamble, Equal Chances, in which the coin decides who gets the 10. Table 8 shows all three gambles.

Table 8: Three gambles.

Unequal

	Heads	Tails
Alex	10	10
Darius	0	0

Equal Chances

	Heads	Tails
Alex	10	0
Darius	0	10

Equal

	Heads	Tails
Alex	5	5
Darius	5	5

(1) Unequal $\sim$ Equal Chances To get from Unequal to Equal Chances, look at each state of nature separately. If the coin lands Heads, the two gambles give the same outcome, $(10, 0)$. If it lands Tails, Unequal gives $(10, 0)$ and Equal Chances gives $(0, 10)$. These are anonymously equivalent, so they are equally good, by Anonymity. So in every state of nature, the two gambles lead to equally good outcomes.

Statewise Dominance then applies both ways round. UNEQUAL is at least as good as EQUAL CHANCES in every state, so UNEQUAL $\succsim$ EQUAL CHANCES. And EQUAL CHANCES is at least as good as UNEQUAL in every state, so EQUAL CHANCES $\succsim$ UNEQUAL. Together, these say that UNEQUAL $\sim$ EQUAL CHANCES.

(2) EQUAL CHANCES $\sim$ EQUAL To get from EQUAL CHANCES to EQUAL, look at each person separately. In EQUAL CHANCES, Alex's expected utility is

$$\tfrac{1}{2} \cdot 10 + \tfrac{1}{2} \cdot 0 = 5.$$

In EQUAL, Alex gets 5 however the coin lands, so his expected utility is also 5. The same goes for Darius: in EQUAL CHANCES, his expected utility is $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 10 = 5$, and in EQUAL it is 5 for sure.

So, by Expected Utility, the two gambles are equally good for Alex, and equally good for Darius. Ex Ante Pareto then applies both ways round: each gamble is at least as good for everyone as the other. So EQUAL CHANCES $\succsim$ EQUAL and EQUAL $\succsim$ EQUAL CHANCES; that is, EQUAL CHANCES $\sim$ EQUAL.

Putting it together. We now have

UNEQUAL $\sim$ EQUAL CHANCES	(Anonymity, Statewise Dominance)
EQUAL CHANCES $\sim$ EQUAL	(Expected Utility, Ex Ante Pareto)
UNEQUAL $\sim$ EQUAL	(Transitivity)

So $(10, 0)$ is exactly as good as $(5, 5)$, which is what Utilitarianism says, since they have the same total.

That is the idea of the argument. If you understand this bit, you understand virtually the full thing. But let's go ahead anyway and do it a bit more carefully in the general case.

The general case makes the same two moves with any number of people, and then uses them to rank any two outcomes.

4.2 The general case

Suppose there are n people. An outcome $X = (x_1, \ldots, x_n)$ is a list of numbers, one per person: x_i is person i's utility in X. We need two definitions.

Definition 2 (Equalisation)**.** The *equalisation* of outcome X, written $\mathsf{E}(X)$, is the outcome in which everyone gets the average of the numbers in X, that is, $(x_1 + \cdots + x_n)/n$.

(The E is for "equalisation"; it's notation I made up which has nothing to do with expectation.)

For example, with three people, Alex, Darius and Philip, let $X = (1, 2, 3)$. The average of 1, 2 and 3 is 2, so $\mathsf{E}(X) = (2, 2, 2)$. In the simple case, the equalisation of $(10, 0)$ is $(5, 5)$, which is EQUAL.

Definition 3 (Lottery-equalisation)**.** A *lottery-equalisation* of outcome X, written $\mathsf{LE}(X)$, is a gamble with n equally likely states of nature (as many states as there are people), such that:

(a) in each state, the outcome is anonymously equivalent to X; and

(b) for each place in the list X, each person gets the number in that place in exactly one state.

There is usually more than one lottery-equalisation of an outcome, but it doesn't matter for the argument which one we pick. Here's a construction which always works: number each person from 1 to n, and likewise number the n equal-probability states from 1 to n. In the first state, we give the first person the wellbeing level the first person gets for sure in X; in the second state, we give them the wellbeing level of the second person; in the third state, the wellbeing level of the third person; and so on. For the second person, we just add one to the number each time, so that they start off at their own level in X, and move along the distribution until, in state n, they wrap around again and receive the wellbeing level that the first person got in X. In general, in the lottery-equalisation of X, person i in state j gets the same wellbeing level as person $i + j - 1$ does in X, counting round: after person n comes person 1 again, so person $n + 1$ is person 1, person $n + 2$ is person 2, and so on.

The second table in Table 9 is built that way, for $X = (1, 2, 3)$. Alex is person 1, so across the three states he gets the levels of persons 1, 2 and 3 in X: 1, then 2, then 3. Darius is person 2, so he starts at his own level, 2, moves on to 3, and then wraps around to 1. Philip starts at 3, and wraps around to 1 and then 2. So in each state, one person gets 1, one person gets 2, and one person gets 3, which means the outcome is anonymously equivalent to X; and across the three states, each person gets each level once. The first table in Table 9 shows X itself as a gamble with the same three states: it gives X in every one of them.

Table 9: $X = (1, 2, 3)$ for sure, and a lottery-equalisation of X. Each state has probability $\frac{1}{3}$.

X

	State 1	State 2	State 3
Alex	1	1	1
Darius	2	2	2
Philip	3	3	3

$\mathsf{LE}(X)$

	State 1	State 2	State 3
Alex	1	2	3
Darius	2	3	1
Philip	3	1	2

In the simple case, Equal Chances is a lottery-equalisation of $(10, 0)$: there are two equally likely states, and Alex and Darius each get 10 in one of them and 0 in the other.

The argument now goes in four steps. The first two are the two moves from the simple case, the third puts them together, and the fourth uses the third to compare any two outcomes.

Step 1: $X \sim \mathsf{LE}(X)$, for any outcome X. Compare X and $\mathsf{LE}(X)$ state by state. In each state, the sure gamble gives the outcome X, and by part (a) of Definition 3, $\mathsf{LE}(X)$ gives an outcome that is anonymously equivalent to X. By Anonymity, that outcome is equally as good as X. So in every state of nature, the outcome of $\mathsf{LE}(X)$ is equally as good as the outcome of the sure gamble.

So each gamble is at least as good as the other in every state. Applying Statewise Dominance both ways round gives $X \succsim \mathsf{LE}(X)$ and $\mathsf{LE}(X) \succsim X$. That is, $X \sim \mathsf{LE}(X)$.

Step 2: $\mathsf{LE}(X) \sim \mathsf{E}(X)$, for any outcome X. Let m be the average of the numbers in X. Take any person i. By part (b) of Definition 3, across the n equally likely states of $\mathsf{LE}(X)$, person i gets the number in each place of X exactly once: that is, they get $x_1, \ldots, x_n$, each in one state. So their expected utility in $\mathsf{LE}(X)$ is

$$\tfrac{1}{n} \cdot x_1 + \cdots + \tfrac{1}{n} \cdot x_n = \frac{x_1 + \cdots + x_n}{n} = m.$$

For $X = (1, 2, 3)$, that is $\frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 2 + \frac{1}{3} \cdot 3 = 2$.

In $\mathsf{E}(X)$, person i gets m for sure, so their expected utility there is also m. So, by Expected Utility, $\mathsf{LE}(X)$ and $\mathsf{E}(X)$ are equally good for person i. Since i was anyone, they are equally good for everyone.

So each gamble is at least as good for everyone as the other. Applying Ex Ante Pareto both ways round gives $\mathsf{LE}(X) \succsim \mathsf{E}(X)$ and $\mathsf{E}(X) \succsim \mathsf{LE}(X)$. That is, $\mathsf{LE}(X) \sim \mathsf{E}(X)$.

Step 3: $X \sim \mathsf{E}(X)$, for any outcome X. By Step 1, $X \sim \mathsf{LE}(X)$. By Step 2, $\mathsf{LE}(X) \sim \mathsf{E}(X)$. So, by Transitivity, $X \sim \mathsf{E}(X)$: every outcome is equally as good as its equalisation. In the simple case, this is the result that $(10, 0) \sim (5, 5)$.

Step 4: comparing any two outcomes. Take any two outcomes X and Y.

Suppose first that X has a greater total than Y. Then X has a greater average than Y, since both totals are divided by the same number of people. In $\mathsf{E}(X)$, each person gets the average of X for sure; in $\mathsf{E}(Y)$, each person gets the average of Y for sure, which is smaller. So, by Expected Utility, $\mathsf{E}(X)$ is better for each

person than $\mathsf{E}(Y)$, and by Ex Ante Pareto, $\mathsf{E}(X) \succ \mathsf{E}(Y)$. Putting this together with Step 3, applied to both X and Y, we have

$$X \sim \mathsf{E}(X) \succ \mathsf{E}(Y) \sim Y,$$

and so, by Transitivity, $X \succ Y$. By the same reasoning, if Y has the greater total, then $Y \succ X$.

Suppose instead that X and Y have the same total. Then they have the same average, so $\mathsf{E}(X)$ and $\mathsf{E}(Y)$ are the very same outcome. By Step 3 and Transitivity, $X \sim \mathsf{E}(X) = \mathsf{E}(Y) \sim Y$, so $X \sim Y$.

So the outcome with the greater total is better, and outcomes with the same total are equally good. That is Harsanyi's aggregation theorem: given Anonymity, Statewise Dominance, Ex Ante Pareto, Expected Utility and Transitivity, with the numbers fitted together as described after the Expected Utility principle,

$$X \succsim Y \quad \text{iff} \quad X \text{ has at least as great a total as } Y.$$

(The totals are totals of the numbers in each outcome: see footnote 2.) This version of the theorem, with Statewise Dominance and Anonymity among its premises, is due to Fleurbaey (2009).[4]

5 Objections

5.1 Can we reject the conclusion?

Go back to the outcomes the proof started from, in Table 6. According to Utilitarianism, Unequal and Equal are equally good, because they have the same total. The apparent unfairness isn't simply that utilitarianism adds everything up. It's that we're indifferent between sharing equally between Alex and Darius, and giving everything to Alex. Intuitively, that's clearly unfair.

If we want to reject the conclusion, we have to reject one of the premises. There's a fairness-based objection to each of the two moves in the proof, aimed at the premises behind that move:

Unequal $\sim$ Equal Chances	Statewise Dominance (and Anonymity)
Equal Chances $\sim$ Equal	Ex Ante Pareto (and Expected Utility)

5.2 Diamond's objection

The first objection is due to Diamond (1967).[5] Table 10 writes the two gambles out side by side, with the outcomes as (Alex, Darius).

Table 10: Diamond's case.

	Heads	Tails
Unequal	$(10, 0)$	$(10, 0)$
Equal Chances	$(10, 0)$	$(0, 10)$

Equal Chances seems better than Unequal, because it's fairer: Darius at least gets a chance at the 10. But Equal Chances is no better than Unequal in either state of nature. If the coin lands Heads, the outcome is the same. If it lands Tails, the outcome is $(0, 10)$ rather than $(10, 0)$, and these are anonymously equivalent, so they are equally good, by Anonymity. So Statewise Dominance says that the two gambles are equally good.

So if Equal Chances really is better, as Diamond thought, then Statewise Dominance is not an appropriate assumption to make about the social welfare ordering. (Diamond (1967: 766) puts his objection in terms

[4] Fleurbaey, Marc (2009). "Two Variants of Harsanyi's Aggregation Theorem". In: *Economics Letters* 105.3, pp. 300–302. See his Theorem 2 (p. 301). His proof differs from the one given here.

[5] Diamond, Peter A. (1967). "Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparison of Utility: Comment". In: *Journal of Political Economy* 75.5, pp. 765–766.

of the sure-thing principle (week 4, section 6.2), which he accepts for individual choice, but not for social choice. The proof needs only Statewise Dominance, not the sure-thing principle.)

5.3 The objection to Ex Ante Pareto

The second objection is aimed at the other move, from Equal Chances to Equal (Table 11).

Table 11: The second move.

Equal Chances

	Heads	Tails
Alex	10	0
Darius	0	10

Equal

	Heads	Tails
Alex	5	5
Darius	5	5

Ex Ante Pareto says these are equally good, since each person's expected utility is 5 either way. But Equal Chances guarantees inequality: however the coin lands, one person is sure to end up worse off than the other.

We might use a lottery to "split" a good that can't be divided. But when the thing can be shared out equally, people usually think the best thing is to share it. Intuitively, Equal Chances is at best an imperfect alternative to sharing.

5.4 Decide as you would with full information

Fleurbaey and Voorhoeve (2013) offer a principle that turns this thought into an argument against Ex Ante Pareto.[6]

> **The Principle of Full Information**: If you know that, whichever state of nature obtains, you would rightly rank X above Y once you knew which state it was, then you should rank X above Y now.

Apply it to Equal Chances and Equal. The objector starts from a judgement about outcomes: that $(5, 5)$ is better than $(10, 0)$, and better than $(0, 10)$. Suppose we knew that the coin would land Heads. Then we would be choosing between $(5, 5)$ and $(10, 0)$, and according to the objector, sharing is better. Suppose instead we knew that it would land Tails. Then we would be choosing between $(5, 5)$ and $(0, 10)$, and sharing is better again. So however the coin lands, once we knew the result we would rank Equal above Equal Chances. By the Principle of Full Information, Equal is better than Equal Chances now, before the coin is tossed. But Ex Ante Pareto says they are equally good.

It's worth noting a couple of things about this argument, though. First, the Principle of Full Information is very similar to Statewise Dominance. Second, it doesn't by itself show that Ex Ante Pareto is false. It only does that if we assume that the conclusion of Harsanyi's theorem is false: that sharing really is better than giving everything to one person. So it isn't obviously a knock-down reply to the utilitarian. It's more like an argument that we should reject one principle rather than the other. And since both Statewise Dominance and Ex Ante Pareto are quite plausible, what we might really want is a reason why one of them in particular is false, rather than an argument from the truth of one, and the falsity of utilitarianism, to the falsity of the other.

[6] Fleurbaey, Marc and Alex Voorhoeve (2013). "Decide As You Would With Full Information! An Argument Against Ex-Ante Pareto". In: *Inequalities in Health: Concepts, Measures, and Ethics*. Ed. by Nir Eyal et al. New York: Oxford University Press. Chap. 8, pp. 113–128. The principle is on p. 120.

6 Utility and wellbeing

6.1 The Harsanyi–Sen–Weymark debate

There is a long-standing debate about what the aggregation theorem shows.[7] One side objects as follows. The theorem shows that what's good overall is a sum of *utilities*. But nothing in the premises connects utility with wellbeing. So the theorem doesn't show utilitarianism, which is a view about the sum of *wellbeing*. (This is the qualification in footnote 2.)

So does the theorem really show utilitarianism? And how might a utilitarian reply to the objection? To answer these questions, we need to be clearer about what the numbers in the proof are.

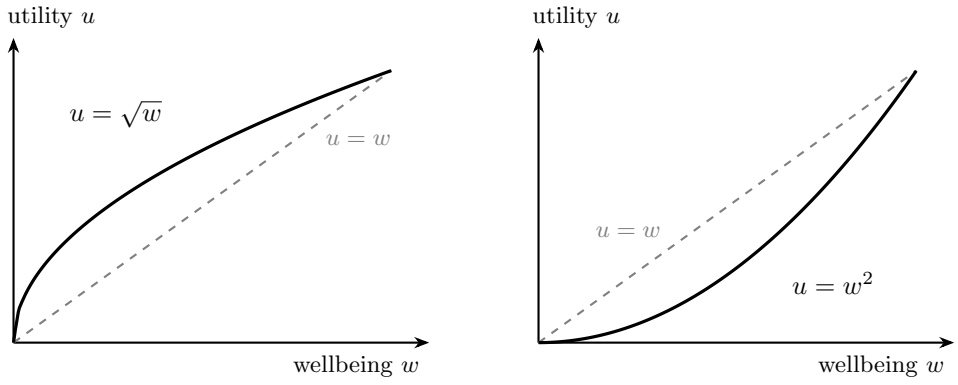
6.2 The numbers are utilities

In the proof, we read the numbers as wellbeing levels. Strictly speaking, they are utilities. By Expected Utility, each person's betterness relation can be represented by a utility function: numbers whose expectation tells us what is better for that person. But, as we saw in section 2.4, a utility function is only fixed up to stretching and shifting. So which utility functions are we using, and how do they relate to wellbeing?

6.3 Utility isn't defined as wellbeing

Utility is a technical notion.[8] Each person's utility function is defined by their betterness relation over prospects: it is whatever assignment of numbers has the right expectations, so that one prospect has a greater expected utility than another exactly when it is better for that person. Wellbeing, by contrast, is how well off someone is.

It's tempting to think that utility just *is* the amount of wellbeing. But nothing in the definition says so. The two can come apart: utility might be a curved function of wellbeing, rather than a straight line. Figure 2 shows two ways it could curve.



Concave: risk averse about wellbeing

Convex: risk seeking about wellbeing

Figure 2: Two ways utility could curve against wellbeing. The dashed line, $u = w$, is for comparison.

6.4 Risk attitudes about wellbeing

Suppose that Darius's utility is the square root of his wellbeing, as in Table 12.

[7] Sen, Amartya (1976). "Welfare Inequalities and Rawlsian Axiomatics". In: *Theory and Decision* 7.4, pp. 243–262; Weymark, John A. (1991). "A Reconsideration of the Harsanyi-Sen Debate on Utilitarianism". In: *Interpersonal Comparisons of Well-Being*. Ed. by Jon Elster and John E. Roemer. Cambridge: Cambridge University Press. Chap. 8, pp. 255–320; Greaves, Hilary (2017). "A Reconsideration of the Harsanyi-Sen-Weymark Debate on Utilitarianism". In: *Utilitas* 29.2, pp. 175–213.

[8] On the relation between utility and wellbeing, see Greaves, Hilary (2015). "Antiprioritarianism". In: *Utilitas* 27.1, pp. 1–42.

Table 12: Darius's utility is $\sqrt{\text{wellbeing}}$.

Wellbeing	0	1	4	9
Darius's utility	0	1	2	3

Compare two prospects for Darius. In the first, he gets wellbeing 4 for sure, which has utility 2. In the second, he has a 50–50 chance of wellbeing 1 or wellbeing 9, and so his expected utility is

$$\tfrac{1}{2} \cdot 1 + \tfrac{1}{2} \cdot 3 = 2.$$

So the two prospects are equally good for Darius, even though the gamble gives him more wellbeing on average: its expected wellbeing is $\frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 9 = 5$, rather than 4. In other words, Darius's betterness relation is *risk averse about wellbeing*: a gamble has to offer him more wellbeing on average than a sure thing, if it is to be equally good for him. That is what the concave (downward-bending) curve in Figure 2 represents. The convex (upward-bending) curve represents the opposite, risk seeking about wellbeing.

But Darius's betterness relation is never risk averse about *utility*: a gamble is always exactly as good for him as a sure thing with the same expected utility. That isn't a fact about Darius. It follows from the definition of utility: expected utility is what tells us what's better for him.

6.5 Same Scale

To compare different people, we use the facts about wellbeing to fit their utility functions to one another, using this constraint:

Same Scale: When two people get the same utility, they are equally well off.

Same Scale only needs comparisons of wellbeing *levels* ("Alex is exactly as well off as Darius"), not comparisons of *amounts* of wellbeing ("Alex is twice as well off as Darius"). With Same Scale in place, the same utility means the same wellbeing, across people. That is also what makes it sensible to apply Anonymity to the numbers: two outcomes whose numbers can be matched up person to person are anonymously equivalent in wellbeing. But it does not follow that twice the utility means twice the wellbeing.

6.6 Same Scale needs the same risk attitude

Can we always meet Same Scale? Only if everyone's betterness relation has the same attitude to risk about wellbeing: in terms of Figure 2, only if everyone's utility follows the same curve.

Suppose, for example, that Alex's betterness relation is risk neutral about wellbeing, so that his utility is a straight line in wellbeing, while Darius's is risk averse, with utility the square root of wellbeing, as before (Table 13).

Table 13: Two people with different risk attitudes about wellbeing.

Wellbeing	0	1	4	9
Alex's utility = wellbeing	0	1	4	9
Darius's utility = $\sqrt{\text{wellbeing}}$	0	1	2	3

Here their utility functions agree at wellbeing 0 and wellbeing 1. But they come apart at 4 and 9. At wellbeing 4, Darius gets utility 2, which is the utility Alex gets at a different level of wellbeing (wellbeing 2); so the same utility no longer means the same wellbeing. Stretching or shifting either function can't fix that: stretching and shifting turn a straight line into a straight line and a curve into a curve (a concave curve stays concave), so they can never make Darius's numbers agree with Alex's at every level of wellbeing.

6.7 Why betterness shares a risk attitude

So is it plausible that everyone's betterness relation has the same attitude to risk? If we were working with preferences, maybe not. Unless rationality requires some particular attitude to risk, a range of risk attitudes might be rationally permissible, and there would be no reason to expect different people's preferences to share one.

But betterness tacitly uses the *correct* attitude to risk. When we ask whether a prospect is better for someone, we are judging what the best balance of risk is for that person, and that doesn't seem to depend so much on the attitudes of the people in question. What's good for you is good for you, regardless of what you prefer (arguably, anyway).

6.8 What the theorem really says

For each person, there are numbers whose *expectation* says what's good for them: their utilities, fitted together by Same Scale. According to the theorem, the *sum* of those same numbers says what's good overall.

"The greatest sum of wellbeing" is exactly right only if betterness is risk neutral about wellbeing, so that utility is a straight line in wellbeing. (Then, by Same Scale, everyone has the same straight line, $u = aw + c$ with $a > 0$, so each outcome's sum of utilities is a times its sum of wellbeing, plus nc; and the two sums rank outcomes the same way.) Otherwise, what the theorem gives us is the greatest sum of utilities. That's the qualification in footnote 2.

6.9 VNM utilitarianism

So even if the premises are true, the theorem's conclusion is not necessarily utilitarianism. Call it *VNM utilitarianism* (VNM: von Neumann–Morgenstern): overall betterness follows the total of utilities, of the specific kind that is linked to wellbeing in the right way.

References

Diamond, Peter A. (1967). "Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparison of Utility: Comment". In: *Journal of Political Economy* 75.5, pp. 765–766.

Fleurbaey, Marc (2009). "Two Variants of Harsanyi's Aggregation Theorem". In: *Economics Letters* 105.3, pp. 300–302.

Fleurbaey, Marc and Alex Voorhoeve (2013). "Decide As You Would With Full Information! An Argument Against Ex-Ante Pareto". In: *Inequalities in Health: Concepts, Measures, and Ethics*. Ed. by Nir Eyal et al. New York: Oxford University Press. Chap. 8, pp. 113–128.

Francis, Tomi (n.d.). "Anonymity, Infinity, and the Separateness of Persons". Unpublished manuscript.

Greaves, Hilary (2015). "Antiprioritarianism". In: *Utilitas* 27.1, pp. 1–42.

– (2017). "A Reconsideration of the Harsanyi-Sen-Weymark Debate on Utilitarianism". In: *Utilitas* 29.2, pp. 175–213.

Gustafsson, Johan E. and Kacper Kowalczyk (2025). "Harsanyi's Utilitarian Theorems without Tears". In: *An Introduction to Utilitarianism*. Ed. by Richard Yetter Chappell, Darius Meissner, and William MacAskill.

Harsanyi, John C. (1955). "Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparisons of Utility". In: *Journal of Political Economy* 63.4, pp. 309–321.

Sen, Amartya (1976). "Welfare Inequalities and Rawlsian Axiomatics". In: *Theory and Decision* 7.4, pp. 243–262.

Weymark, John A. (1991). "A Reconsideration of the Harsanyi-Sen Debate on Utilitarianism". In: *Interpersonal Comparisons of Well-Being*. Ed. by Jon Elster and John E. Roemer. Cambridge: Cambridge University Press. Chap. 8, pp. 255–320.